

```
[root@DELL-RnD-India tracing]# pwd
/sys/kernel/debug/tracing
[root@DELL-RnD-India tracing]# cat trace_options
print-parent
nosym-offset
nosym-addr
noverbose
noraw
nohex
nobin
noblock
nostacktrace
trace_printk
noitrace_preempt
nobraunch
annotate
```

Figure 1: Trace options

```
[root@DELL-RnD-India ~]# ./traceprocess.sh ls
Desktop Music Templates init kexec.log my.syscall.c traceprocess.sh
Documents Pictures Videos install.log miniconf.log rsbuild
Downloads Public booting install.log syslog mount shutdown
[root@DELL-RnD-India ~]# cat /sys/kernel/debug/tracing/set_ftrace_pid
1078
[root@DELL-RnD-India ~]# cat /sys/kernel/debug/tracing/trace | grep ls -l | head -10
ls-10376 [002] 292426.516646: mutex_unlock <-tracing set tracer
ls-10376 [002] 292426.516646: fnotify modify <-fs write
ls-10376 [002] 292426.516647: fnotify parent <-fnotify modify
ls-10376 [002] 292426.516647: fnotify <-fnotify modify
ls-10376 [002] 292426.516648: sru read lock <-fnotify
ls-10376 [002] 292426.516648: sru read unlock <-fnotify
ls-10376 [002] 292426.516648: fput light <-fs write
ls-10376 [002] 292426.516649: audit syscall exit <-sysret audit
ls-10376 [002] 292426.516650: audit get context <-audit syscall exit
ls-10376 [002] 292426.516651: audit free names <-audit syscall exit
[root@DELL-RnD-India ~]# > /sys/kernel/debug/tracing/set_ftrace_pid
pid
[root@DELL-RnD-India ~]#
```

Figure 2: Executing traceprocess.sh and viewing trace output

```
[root@DELL-RnD-India ~]# echo function_graph > /sys/kernel/debug/tracing/current_tracer
[root@DELL-RnD-India ~]# cat /sys/kernel/debug/tracing/current_tracer
function_graph
[root@DELL-RnD-India ~]# head -20 /sys/kernel/debug/tracing/trace
# Tracer: function_graph

#
# DURATION      FUNCTION CALLS
#-----
smpt_apic_timer_interrupt() {
#
# 0.719 us      native_apic_mem_write();
# 0.707 us      native_apic_mem_write();
# 0.751 us      native_apic_mem_write();
# 0.312 us      exit_j0fe();
# 0.314 us      exit_j0fe();
# 0.335 us      irq_enter() {
#               irq_enter() {
#               irq_enter() {
#               rcu_irq_enter();
0.311 us      rcu_irq_enter();
[root@DELL-RnD-India ~]#
```

Figure 3: Trace output of function graph tracer

```
# execute the process
exec $*
```

Once tracing is complete, you need to clear the `set_ftrace_pid` file, for which you can use the following command:

```
> set_ftrace_pid
```

Function graph tracer

The function graph tracer tracks the entry and exit of a function, and is quite useful to track

its execution time. Functions with a duration of over 10 microseconds are marked with a “+”, and those over 100 microseconds with “!”. To enable the function graph tracer, use `echo function_graph > current_tracer`. The sample output is as shown in Figure 3. There are a lot of tracers; the entire list is in [linux/Documentation/trace/trace.txt](#). The tracers are enabled or disabled by echoing the tracer name into the `current_tracer` file.

```
[root@DELL-RnD-India tracing]# cat available_filter_functions | grep vmlalloc
vmlalloc_sync all
ls vmlalloc or module addr
vmlalloc to page
vmlalloc to pin
vmlalloc to pin
renap vmlalloc_range
vmlalloc node
vmlalloc area node
vmlalloc area
vmlalloc 32 user
vmlalloc 32
vmlalloc exec
vmlalloc node
vmlalloc user
vmlalloc
vmlalloc
vmlalloc open
get vmlalloc info
vmlalloc to mfn
[root@DELL-RnD-India tracing]# echo vmlalloc * > set_ftrace_filter
[root@DELL-RnD-India tracing]# cat set_ftrace_filter
vmlalloc_sync all
vmlalloc to page
vmlalloc to pin
vmlalloc to pin
vmlalloc 32 user
vmlalloc 32
vmlalloc exec
vmlalloc node
vmlalloc user
vmlalloc open
vmlalloc to mfn
[root@DELL-RnD-India tracing]#
```

Figure 4: Listing filter functions, using with wild-cards

```
[root@DELL-RnD-India tracing]# cat available_events | head -10
kvmnm: kvm_mmu_pageable_walk
kvmnm: kvm_mmu_paging_element
kvmnm: kvm_mmu_set_accessed_bit
kvmnm: kvm_mmu_set_dirty_bit
kvmnm: kvm_mmu_walker_error
kvmnm: kvm_mmu_get_page
kvmnm: kvm_mmu_sync_page
kvmnm: kvm_mmu_unsync_page
kvmnm: kvm_mmu_prepare_zap_page
kvm: kvm_entry
[root@DELL-RnD-India tracing]# echo sys_enter_nice >> set_event
syscallsys: sys_enter_nice
[root@DELL-RnD-India tracing]# echo 'sys_enter_nice' >> set_event
[root@DELL-RnD-India tracing]# cat set_event
[root@DELL-RnD-India tracing]#
```

Figure 5: Available tracing events, setting and ‘unsetting’ them.

Dynamic tracing

We can easily get inundated with the amount of data the function tracer throws at us. There is a dynamic way to filter just the functions we need, and eliminate those that we don’t need: to specify them in the file `set_ftrace_filter`. (First find the function(s) you want, from the `available_filter_functions` file.) See Figure 4 for an example of dynamic tracing. As you can see, you can even use wild-cards for the functions’ names. I used all the `vmlalloc_` functions, and set them with: `echo vmlalloc_* > set_ftrace_filter`.

Event tracing

Tracing can also be triggered when particular events happen on the system. Available system events are found in the file `available_events`:

```
[root@DELL-RnD-India tracing]# cat available_events | head -10
kvmnm: kvm_mmu_pageable_walk
kvmnm: kvm_mmu_paging_element
kvmnm: kvm_mmu_set_accessed_bit
kvmnm: kvm_mmu_set_dirty_bit
kvmnm: kvm_mmu_walker_error
kvmnm: kvm_mmu_get_page
kvmnm: kvm_mmu_sync_page
```